

AECCS Final Presentation

Automated GDPR compliance analysis and PET evaluation for cookie consent systems.

CS475 Privacy-Enhancing Technologies

Research Question

- Do cookie consent systems actually protect users from non-essential tracking?
- How often do banners rely on dark patterns?
- Which PETs reduce exposure most effectively when consent systems fail?

What Was Built

- Crawler with no-interaction, accept, and reject states.
- Tracker classification using filter lists and heuristics.
- Dark-pattern detection and GDPR-oriented compliance scoring.
- Browser PET evaluation, CMP comparison, and differential privacy reporting.
- Visualization and HTML reporting pipeline with provenance checks.

Current Repository Status

Area	Status
Software pipeline	Implemented
Mock/demo dataset	Reproducible
Real study list	Prepared with 100 domains
Real crawl outputs	Not yet executed
Final empirical results	Pending real study

Current Demo Findings

- 84.2% pre-consent tracker violations in the mock dataset.
- 75.0% of mock sites show at least one dark pattern.
- Average mock compliance score is 50.8 / 100.
- Brave Shields is the strongest browser PET in the mock comparison.
- Didomi is the strongest CMP in the mock comparison.

These values are demonstration outputs from data/mock/, not final real-study claims.

Final Actions Before Submission

- Run the real crawl and PET evaluation under `data/real/`.
- Regenerate every figure and report from real processed data only.
- Replace the demo findings in the report and slides with real results and citations.